

Appendix I - 8 key principles: explanation boxes

The 8 key principles of the SU-EATABLE DIET – THE WHY

The following boxes report the scientific evidence in support of the 8 principles on which the SU-EATABLE LIFE diet is based. Environmental impact is specifically addressed with reference to the two main indicators of the SU-EATABLE Life Project: CO₂ emissions equivalents, and water quality and quantity. The amount of CO₂ emitted per kg or litre of food product is referred to as Carbon Footprint (question box n°2), or is reported as CO₂ equivalents (question box n°4). The amount of water used per kg or litre of food product is indicated as Water Footprint – WF (question box n°3).

1. Champion plant-based food in your diet

Environment: Plant-based food emits on average significantly less CO₂ equivalents (0.2 to 2.7 kg CO₂-eq/kg) compared to animal-based food (3.5 to 28 kg CO₂-eq/kg), with the exception of milk and eggs, which are in the range of plant-based food (Clune et al. 2017, Poore and Nemecek 2018, Willet et al. 2019). Large savings in WF with dietary choices can be achieved depending on the weekly menu. The WF of vegetables, grains and fruit is about 25 times lower when compared to beef; as a protein source, legumes have a WF from 2 to 4 times higher than the beef WF and similar to the WF of chicken and eggs (Mekonnen, M.M. and Hoekstra, A.Y., 2010). The optimal daily diet [-composition-] for health and the environment which could help the planet to stay within its sustainable resource use should consist, by volume, of approximately half a plate of vegetables and fruits, and a significant source of protein should still be provided by legumes, nuts and cereals (Willet et al. 2019). To date, in industrialized countries, people tend to consume too much sugar, oils and fats, red meat as well as milk and cheese combined, while not enough fruit and vegetables (Vanham et al 2018, Willet et al. 2019).

Health: According to the World Health Organization, a healthy diet includes: fruit, vegetables, legumes, nuts and whole grains, with at least 400 g (i.e. five portions) of fruit and vegetables per day (WHO, 2018). Emphasizing nutrient-dense, plant-based foods such as fruit, vegetables, legumes and nuts, is associated with significantly better health outcomes (Tuso et al. 2013; Satija et al., 2016; BDA, 2017; Satija et al., 2017; Toumpanakis et al. 2018).